# VEERMATA JIJABAI TECHNOLOGICAL INSTITUTE

Matunga, Mumbai-400 019

**Autonomous Institute affiliated to University of Mumbai**

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| EXAMINATION | **Mid Sem Exam May 2025** | DATE OF EXAM | **17/03/2025** |
| SEMESTER & PROGRAM | **Sem - II, First Year MCA** | TIME | **11.15 am to 12.45 pm** |
| TIME ALLOWED | **1&1/2 HRS.** | MARKS | **40** |
| COURSE NAME – (CODE) | **DESIGN AND ANALYSIS OF ALGORITHM – (R5MC5022T )** | | |

Instructions
1. All questions carry equal marks.
2. Figures to the right indicate full marks.
3. Assume suitable data if required

Q.1 a. A machine needs a minimum of 400 sec to sort 1500 elements by Quick sort. The minimum time needed to sort 300 elements will be? (4) CO3

b. What is asymptotic notation? Explain Big O (Big - Oh) and Big- Θ (Big- Theta) asymptotic notation. (6) CO1

Q.2 a. What is stable sorting and give example of stable and unstable sorting algorithms. (4) CO1

b. Write the pseudocode for selection sort using recursion. (6) CO2

Q.3 a. Write the time complexity of the following algorithm: (4) CO3

(i)
```
function(int n)
{
    if (n==1)
        return;
    for (int i=1; i<=n; i++)
    {
        for (int j=1; j<=n; j++)
        {
            printf("*");
            break;
        }
    }
}
```

(ii)
```
int a = 0, i = N;
while (i > 0) {
  a += i;
  i /= 2; }
```

b. Sort the following elements using merge sort method. (6) CO2
45, 12, 89, 33, 77, 23, 9, 51, 60, 19, 40

4 a. What is the time complexity of performing binary search on a sorted linked list? (2) CO1

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|---|---|---|---|
| EXAMINATION | Mid Semester Examination March 2025 | DATE OF EXAM | Wednesday 20.03.2025 |
| SEMESTER & PROGRAM | Sem-II, First Year MCA | TIME | 11.15 am 12.45pm |
| TIME ALLOWED | **1 hour 30 minutes** | MARKS | **40** |
| COURSE NAME – (CODE) | Professional Communication ( R5MC5027T ) | | |

**Instructions**

1. All questions carry ten marks each.
2. Figures to the right indicate full marks.
3. Choose any four questions out of the five and answer.

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|---|---|---|---|
| **Q.1** | | Emma, the Marketing Manager at XYZ Corp, sent an email outlining campaign details for a product launch to her team. However, communication issues arose and the team did not receive the email due to server issues. As a result, the team faced delays in the timeline of launching the campaign. | |
| | a. | Based on the following case study, create a diagram of the process of communication and label the sender, receiver and channel. | **(5)** |
| | b. | Explain each aspect of the process of the process of communication and why feedback is an important part in the process of communication. | **(5)** |
| | | | |
| **Q.2** | a. | Mark relevant arrows on the following sentences to indicate whether it is a rising, falling or rising-falling intonation along with an explanation of all three types.<br>*i) Are you coming to the meeting?*<br>*ii) The report is due tomorrow*<br>*iii) Do you want tea or coffee?* | **(6)** |
| | b. | Why is intonation and modulation important while speaking? Give four reasons. | **(4)** |
| | | | |
| **Q.3** | a. | Effective oral presentation skills are crucial in communicating your ideas. Explain the three stages of thinking, planning and talking in detail. | **(5)** |

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|---|---|---|---|
| | b. | You have been asked to create an infographic to summarize complex data for a business report. Describe three key design principles you would follow to ensure clarity and effectiveness. | (5) |
| Q.4 | a. | What are the four channels of communication in an organization? Explain what they are and how they function. | (5) |
| | b. | Imagine you are creating a diagrammatic representation of the channels of communication in VJTI. Using the words given below, create suitable diagrams to illustrate examples of the four channels.<br>*HoD, Student, Professor, Dean, Director, Deputy Director, Training and Placement Officer, Registrar* | (5) |
| Q.5 | | At BrightEdge Marketing in Bangalore, Meera, a junior executive, presented a product proposal to a French client but used complex industry jargon, making it hard for them to follow. As the meeting progressed, there were audio lags and background noise from her crowded office which led to a lot of missed details and miscommunication. During the course of her presentation, she also expected immediate feedback, but her French client had a more formal approach and waited till the end to raise his concerns. Nervous and unsure about how to react, she gave vague responses, reducing the client's confidence. After the meeting ended her boss was unhappy with her performance and called her to discuss what went wrong. | |
| | a. | After reading the following case study, note down the barriers in detail, with reasons. | (5) |
| | b. | Write down possible solutions for each barrier. What should Meera/ her company do to solve these issues? | (5) |

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| *EXAMINATION* | Mid Semester Examination March 2025 | DATE OF EXAM | 19/03/2025 |
| *SEMESTER & PROGRAM* | Sem-II, First Year MCA | TIME | 11.15 am – 12.45pm |
| *TIME ALLOWED* | **1 1/2 HR** | MARKS | **40** |
| *COURSE NAME – (CODE)* | Cloud Computing – (R5MC5113S) | | |

*Instructions*

1. All questions carry equal marks.
2. Figures to the right indicate full marks.
3. Assume suitable data if required.

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|---|---|---|---|
| Q.1 | a. | Define cloud computing? Elaborate on the various 5 essential attributes of Cloud Computing? | (5) |
| | b. | Which are the different 7 steps for migration to cloud? Explain in detail. | (7) |
| | c. | What is a virtualization? Explain hardware assisted virtualization in detail with diagram. | (5) |
| | d. | Differentiate between HTML and XML. Give advantages of XML. | (3) |
| | | | |
| Q.2 | a. | Differentiate between paravirtualization and full virtualization. (At least valid 5 points) | (5) |
| | b. | What are the different PaaS characteristics? Which are the various applications where PaaS may not be useful? | (5) |
| | c. | What are the different distributed computing paradigms? Explain in brief. | (5) |
| | d. | Explain trap-and-emulate hypervisor design technique in detail with diagram. | (5) |
| | | **All the best** | |

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| | | | essential attributes | (5) |
| | c | What is a virtualization? Explain hardware assisted virtualization in detail with diagram | | (7) |
| | d | Differentiate between HTML and XML. Give advantages of XML | | (5) |

**VEERMATA JIJABAI TECHNOLOGICAL INSTITUTE (VJTI)**
**[Central Technological Institute, Maharashtra State]**
**Matunga, Mumbai-400 019**

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| EXAMINATION | Mid Semester Examination March 2025 | DATE OF EXAM | Friday 21.03.2025 |
| SEMESTER & PROGRAM | 2nd & MCA | TIME | 11.15 AM to 12.45 PM |
| TIME ALLOWED | 1.5 Hrs | MARKS | 40 |
| COURSE (Course Code) | **R5MC5023S Mathematical and Statistical Foundation 2** | | |

**Instructions: All questions are Compulsory**

| Q. No | Sub-ques. | Question | Marks | CO |
|---|---|---|---|---|
| Q.1 | a | When a relation becomes a equivalence relation? Is congruent (≅)a equivalence relation? | 5M | CO1 |
| | b | Find quadratic residue mod 11. Also find quadratic non residue | 5M | CO1 |
| Q.2 | a | What is cyclic group? Is cyclic group always an abelian group? If yes, Why? If not, Why? | 5M | CO2 |
| | b | Does $Z_p^*$ over multiplication is a group? Is it an abelian group? | 5M | CO1 |
| Q.3 | a | If N=2020 then φ(N) is? | 5M | CO3 |
| | b | If p=11 and q=7 then then find multiplicative inverse of 13 modulo φ(p*q) | 5M | CO1 |
| Q.4 | a | Is ($Z_p$ ,+ addition,* multiplication) is a field? Check for p=17 | 5M | CO3 |
| | b | What are the properties a set should satisfy to become commutative ring. | 5M | CO1 |

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